

Variables & Game HUD Text

Day 02

Lesson 02

Nora XIE

Vibe Coding Bootcamp

Phase 1 — I can make the computer do things Time: 60–75 min Mode: Self-study

Today's goals:

- ★ Store data in variables
- ★ Build a 3-line player status screen
- ★ Use f-strings for readable output



Quest map: Day 02 of 22

Hook — why today matters

In games you see name, level, and HP. Variables are boxes that hold those values.

Big idea



Learn

A **variable** has a name and a value. `f'...'` strings insert values into text. Good names: `player_hp`, not `x`.



AI rules today

AI allowed in **Investigate** only: ask why an f-string shows a number. No full-file generation.



Demo code

Folder: day02/

```
1 player_name = "Nora"
2 level = 8
3 hp = 42
4 print(f"=== {player_name} ===")
5 print(f"Level {level} | HP {hp}")
```

Guided practice

Try it

Change `hp` to 0. Predict the screen, then run.

Try it

Add a line `mp = 10` and show MP in the HUD.

Assignments

Assignment

Work in: `day02/assignment/`

A — Predict (write each answer before running a test file)

1. `name="A"` then `print(f"Hi {name}")` → _____
2. `print(3*4)` → _____
3. `hp=50` then `print(f"HP:{hp}")` → _____

B — Debug `assignment/buggy_player.py` — fix the f-string.

C — Level up `assignment/player_card.py`: name, level, HP, MP (4 lines output).

D — Explain Why is `player_hp` better than `x`?

Self-study checklist

- ☐ Re-read **Learn**; predict outputs (10 min)
- ☐ Assignments A–D (35–45 min)
- ☐ Run your program 3 times with different inputs
- ☐ 3-bullet “what I tried” note
- ☐ Stuck 15 min → C-R-A-V (`../shared/prompts.md`)

Reflection

1. Easiest vs hardest today?
2. One line you can teach a friend.

Tomorrow: Input from the keyboard and math.